Nuclear and Particle Physics
Assignment 2
Group EE2

Quantum description of scattering theory:

Solid Angle

Description: Solid angle defines the three-dimensional angular region into which particles are scattered. It is analogous to angle in 2D but extended to spherical geometry.

$$d\Omega = \sin\theta \, d\theta \, d\phi$$

Total solid angle: The complete angular space around a point is:

$$\int d\Omega = \int_0^{\pi} \sin\theta d\theta \int_0^{2\pi} d\phi = 4\pi$$

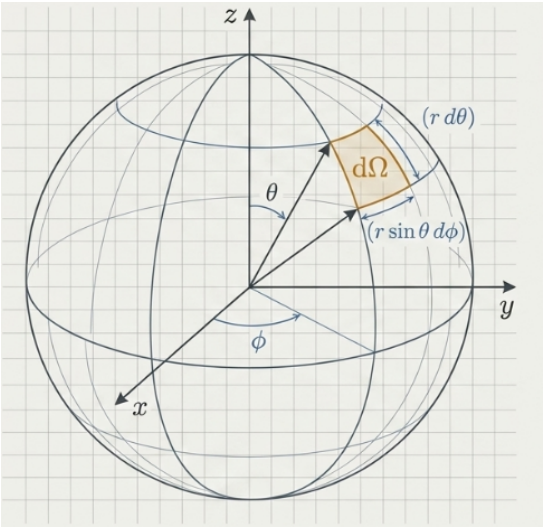


Figure 1: Solid angle element in spherical coordinates

Scattering Cross Section

Description: Cross section quantifies the probability of interaction between incident particles and the target.

Differential cross section: It represents scattering per unit solid angle in a given direction.

$$\frac{d\sigma}{d\Omega}(\theta, \phi)$$

Total cross section: It gives the total scattering over all directions.

$$\sigma = \int \frac{d\sigma}{d\Omega}\, d\Omega$$

Wave Description

Description: In quantum mechanics, scattering is described using wave functions of particles before and after interaction.

Incident wave: A plane wave representing particles moving in a fixed direction.

$$\psi_i \sim e^{ikz}$$

Scattered wave: A spherical wave spreading outward from the scattering center.

$$\psi_s \sim \frac{e^{ikr}}{r} f(\theta, \phi)$$

Scattering amplitude: Determines angular distribution and strength of scattering.

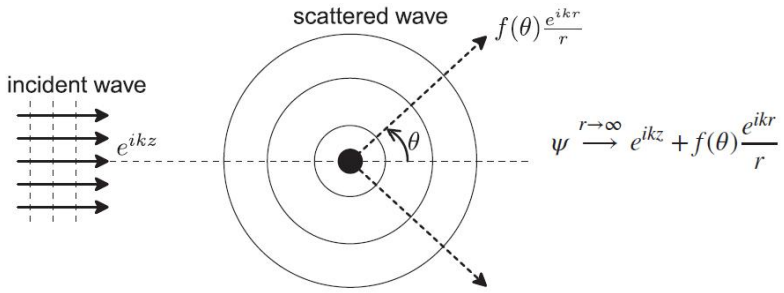


Figure 2: Incident plane wave and scattered spherical wave

Current Density

Description: Current density represents the flow of probability associated with particle motion.

Initial current density:

$$J_i = \left(\frac{\hbar k}{\mu}\right) |\psi_i|^2$$

Scattered current density:

$$J_s = \left(\frac{\hbar k}{\mu}\right) \left| \frac{e^{ikr}}{r} f(\theta, \phi) \right|^2$$

Relation for Cross Section

Description: The differential cross section is obtained by comparing scattered flux to incident flux.

$$\frac{d\sigma}{d\Omega} = \frac{J_s}{J_i} = |f(\theta, \phi)|^2$$

Solution of Schrödinger Equation

Description: The scattering amplitude is expressed as a sum of contributions from different angular momentum states (partial waves).

$$f(\theta) = \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} P_l(\cos\theta) \sin\delta_l$$

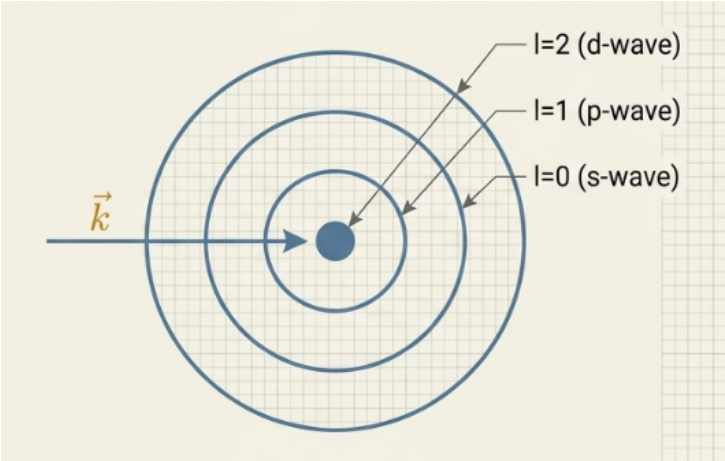


Figure 3: Partial wave contributions for different angular momentum states ($l = 0, 1, 2$)

Total Cross Section

Description: Total scattering is obtained by integrating over all solid angles and summing contributions from all partial waves.

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

$$\sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l$$

Angular Momentum

Description: Angular momentum determines which partial waves contribute significantly to scattering.

Relation:

$$\ell\hbar = r \times p = \hbar k r$$

$$\ell = rk$$

Condition: For nuclear scale ($r \sim 5$ fm), if

$$rk \ll 1 \Rightarrow \ell = 0$$

Only s-wave ($\ell = 0$) contributes significantly.

Low Energy Approximation

Description: At low energies, higher angular momentum contributions are negligible.

$$\sigma \approx \frac{4\pi}{k^2} \sin^2 \delta_0$$

For small k:

$$\delta_0 \approx ka$$

$$\sigma = 4\pi a^2$$

where a is the scattering length.

Hard Sphere Scattering

Description: In this approximation, the target behaves like a rigid sphere.

$$\sigma = 4\pi a^2$$

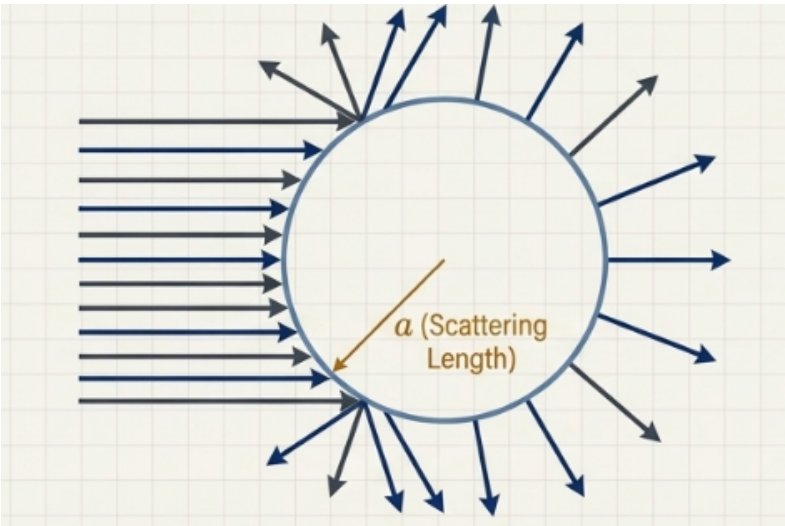


Figure 4: Hard sphere scattering with isotropic distribution over 4π

DSAI+MSME

Radioactive Decay

1 Radioactive Decay

Radioactive elements undergo spontaneous emission of radiation:

- α particles (Helium nucleus)
- β particles (Electron)
- γ rays (High energy photons with high frequency higher than x-rays)

1.1 Examples

Alpha decay:

$$^{238}_{92}U \rightarrow ^{234}_{90}Th + ^{4}_{2}He$$

Beta decay:

$$^{14}_{6}C \rightarrow ^{14}_{7}N + e^{-}$$

Gamma decay:

$$^{87}_{38}Sr^{*} \rightarrow ^{87}_{38}Sr + \gamma$$

2 Radioactive Decay Law

$$N(t) = N_0 e^{-\lambda t}$$

Where:

- $N(t)$ = number of atoms at time t
- N_0 = initial number of atoms
- λ = decay constant

3 Step-by-Step Derivation

Step 1: Basic assumption

Rate of decay is proportional to number of atoms:

$$\frac{dN}{dt} \propto N$$

Step 2: Introduce constant

$$\frac{dN}{dt} = -\lambda N$$

(Negative sign indicates decay)

Step 3: Separate variables

$$\frac{dN}{N} = -\lambda dt$$

Step 4: Integrate both sides

$$\int_{N_0}^{N(t)} \frac{dN}{N} = \int_0^t -\lambda dt$$

Step 5: Solve integrals

$$\ln N(t) - \ln N_0 = -\lambda t$$

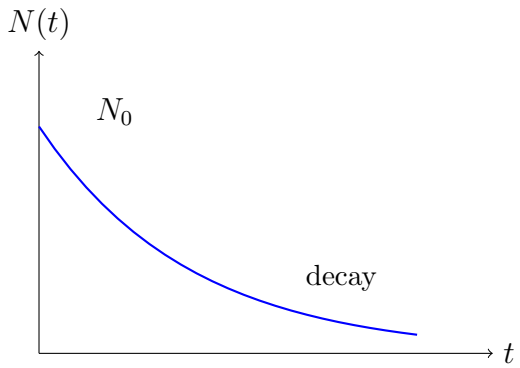
Step 6: Simplify

$$\ln\left(\frac{N(t)}{N_0}\right) = -\lambda t$$

Step 7: Exponentiate

$$\frac{N(t)}{N_0} = e^{-\lambda t}$$

4 Exponential Decay Graph



5 Quantum Description

5.1 Stationary State

$$\psi(t) = e^{-iEt/\hbar}$$

5.2 Decaying State

$$\psi(t) = e^{-iE_0 t/\hbar} \cdot e^{-\Gamma t/2}$$

Interpretation:

- First term → oscillation (energy)
- Second term → decay

6 Probability of Decay

$$|\psi(t)|^2 \propto e^{-\Gamma t}$$

Decay in quantum mechanics also follows exponential law

7 Energy Representation (Fourier Transform)

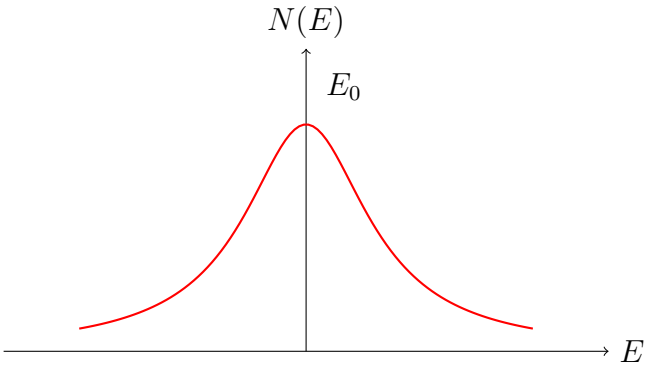
$$\psi(E) \propto \int \psi(t) e^{iEt} dt$$

$$\psi(E) \propto \frac{1}{(E - E_0) + i\Gamma/2}$$

8 Energy Distribution

$$N(E) \propto \frac{1}{(E - E_0)^2 + \Gamma^2/4}$$

9 Lorentzian (Breit-Wigner) Graph



10 Key Insights

- Radioactive decay is exponential in time
- Quantum decay introduces width Γ
- Energy distribution is Lorentzian
- Larger $\Gamma \rightarrow$ faster decay $\rightarrow$ wider peak

Discovery of the Neutrino Particle & Accelerators

Lecture Notes — Nuclear & Particle Physics

Group CS2

1 Discovery of the Neutrino Particle (ν)

Historical Timeline

1896–1900 Becquerel (Nuclear Physicist)
Discovered radioactive uranium salts. Using magnetic-field deflection and measuring the charge-to-mass ratio, Becquerel confirmed that β-rays are flows of *electrons*.

1914–1927 Chadwick et al.
Experimentally observed a *continuous* β-ray energy spectrum — a result that could not be explained by simple two-body decay.

The Problem with Two-Body β-Decay

If β-decay were a two-body process,

$$n \longrightarrow p + e^{-},$$

momentum conservation in the centre-of-mass frame would require the electron to carry a *fixed* momentum $|\vec{p}_{\mathrm{cm}}|$, producing a discrete, *sharp* endpoint energy — **not** the continuous spectrum that was observed.

Pauli's Resolution (1930)

Pauli proposed a third, undetected particle $\bar{\nu}_e$ to restore energy–momentum conservation:

$$\boxed{n \longrightarrow p + e^- + \bar{\nu}_e}$$

The momenta $\vec{p}_e$, $\vec{p}_p$, and $\vec{p}_{\bar{\nu}}$ are now shared among **three** final-state particles, allowing a *continuous* electron spectrum with energies up to (and slightly above) the classical endpoint energy.

2 Accelerators

Linear Accelerator

Single Electrode

A charged particle (charge e, mass m) accelerated through potential V gains kinetic energy:

$$\tfrac{1}{2}\,mv_1^2 = eV \qquad \Longrightarrow \qquad \boxed{v_1 = \sqrt{\frac{2eV}{m}}}$$

n Linear Electrodes

After passing through n successive electrodes (each at potential V):

$$\tfrac{1}{2}\,mv_n^2 = n\,eV \qquad \Longrightarrow \qquad \boxed{v_n = \sqrt{\frac{2n\,eV}{m}} = \sqrt{n}\;v_1}$$

By increasing n, the final velocity of the particle increases as $\sqrt{n}$.

Limitation: A long evacuated (vacuum) drift tube is required, making the apparatus physically large and expensive.

Cyclotron

A cyclotron accelerates a charged particle (charge e, mass m) using two D-shaped electrodes (*dees* D_1 and D_2) immersed in a uniform magnetic field B.

Equation of Motion

The magnetic (Lorentz) force provides the centripetal force:

$$\frac{mv^2}{r} = evB \qquad \Longrightarrow \qquad \boxed{r = \frac{mv}{eB}}$$

Cyclotron (Angular) Frequency

$$\boxed{\omega = \frac{v}{r} = \frac{eB}{m}}$$

This is the *cyclotron angular frequency*: it is **independent of the particle's speed**, which is the key property that makes resonance acceleration possible.

Cyclotron Period

$$T_B = \frac{2\pi}{\omega} = \frac{2\pi m}{eB}$$

Maximum Energy

Let n be the total number of dee-gap crossings and $r_{\max}$ the maximum orbital radius (set by the physical size of the dees):

$$\boxed{E_{\max} = \tfrac{1}{2}\, m v_{\max}^2 = n\, eV = \frac{B^2 e^2 r_{\max}^2}{2m}}$$

Diagram

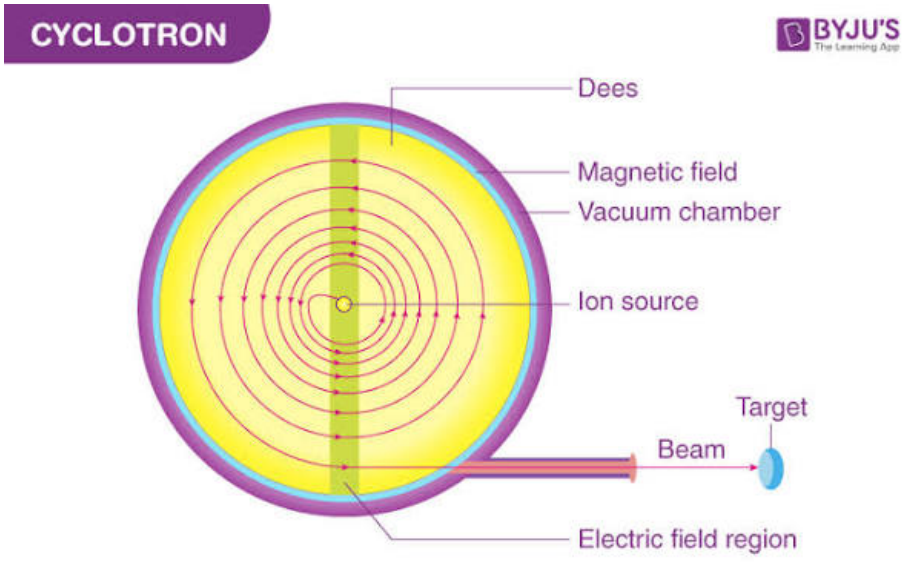


Figure 1: Schematic of a cyclotron showing the two dees D_1 and D_2, the alternating electric field in the gap, and the spiral path of the accelerating particle in the magnetic field B.

ME 2

PHL509 - Assignment2

April 13, 2026

1 Nuclear Density distribution $\rho(r)$:-

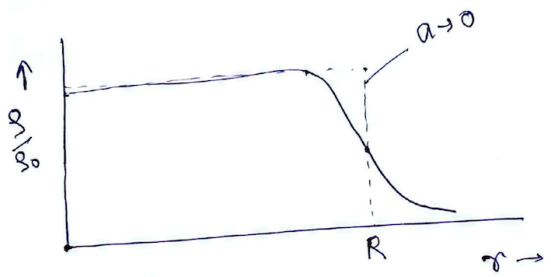
$$\rho(r) = \frac{\rho_0}{\exp\left(\frac{r-R}{a}\right) + 1} \tag{1}$$

$\mathrm{R} = R_o A^{1/3}$ with $R_o \approx 1.3\,\mathrm{fm}$ $1\,\mathrm{fm} = 10^{-15}\,\mathrm{m}$

$$A = N + Z$$

where

- A = mass number
- N = neutron number
- Z = proton number



cross section for structure less nuclei $\rightarrow \sigma_{Mott}$

cross section of actual nuclei $\rightarrow$

$$\sigma = \sigma[F(q)]^2 \tag{2}$$

where form factor

$$F(q) = \frac{4\pi}{q} \int_0^\infty \rho(r) \sin(qr)\, r\, dr \tag{3}$$

$$q = \frac{2E}{\hbar c} \sin\frac{\theta}{2} = \frac{2}{\lambda} \sin\frac{\theta}{2} \tag{4}$$

$$\lambda = \frac{\hbar c}{E} \tag{5}$$

is reduced de-broglie wavelength of electron (probing nuclie).

Normalization condition of charge density

$$\int_0^\infty \rho(r)\, 4\pi r^2\, dr = Ze \tag{6}$$

For $qr \ll 1$ or $\dfrac{r}{\lambda} \ll 1$ (means $\lambda \gg$ nuclear scale r)

$$\sin(qr) = qr - \frac{1}{6} q^3 r^3 + \ldots$$

then Equation ③ become

$$F(q) = \frac{4\pi}{q} \int_0^\infty \rho(r) \left[qr - \frac{1}{6} q^3 r^3 + \ldots \right] r\, dr$$

from the relation $e^{i\theta} = \cos\theta + i \sin\theta$ in binomial expansion.

$$F(q) = \int \rho(r) 4\pi r^2 dr - \frac{1}{6} q^2 \int r^2 \rho(r) 4\pi r^2 dr + \ldots$$

$$F(q) = Ze \left[1 - \frac{1}{6} q^2 \langle r^2 \rangle + \ldots \right]$$

Total electric charge Ze can be defined as

$$Ze = \lim_{q \to 0} F(q)$$

and mean square charge radius can be defined as

$$\langle r^2 \rangle = -6 \left(\frac{dF}{dq^2} \right)_{q^2=0}$$

